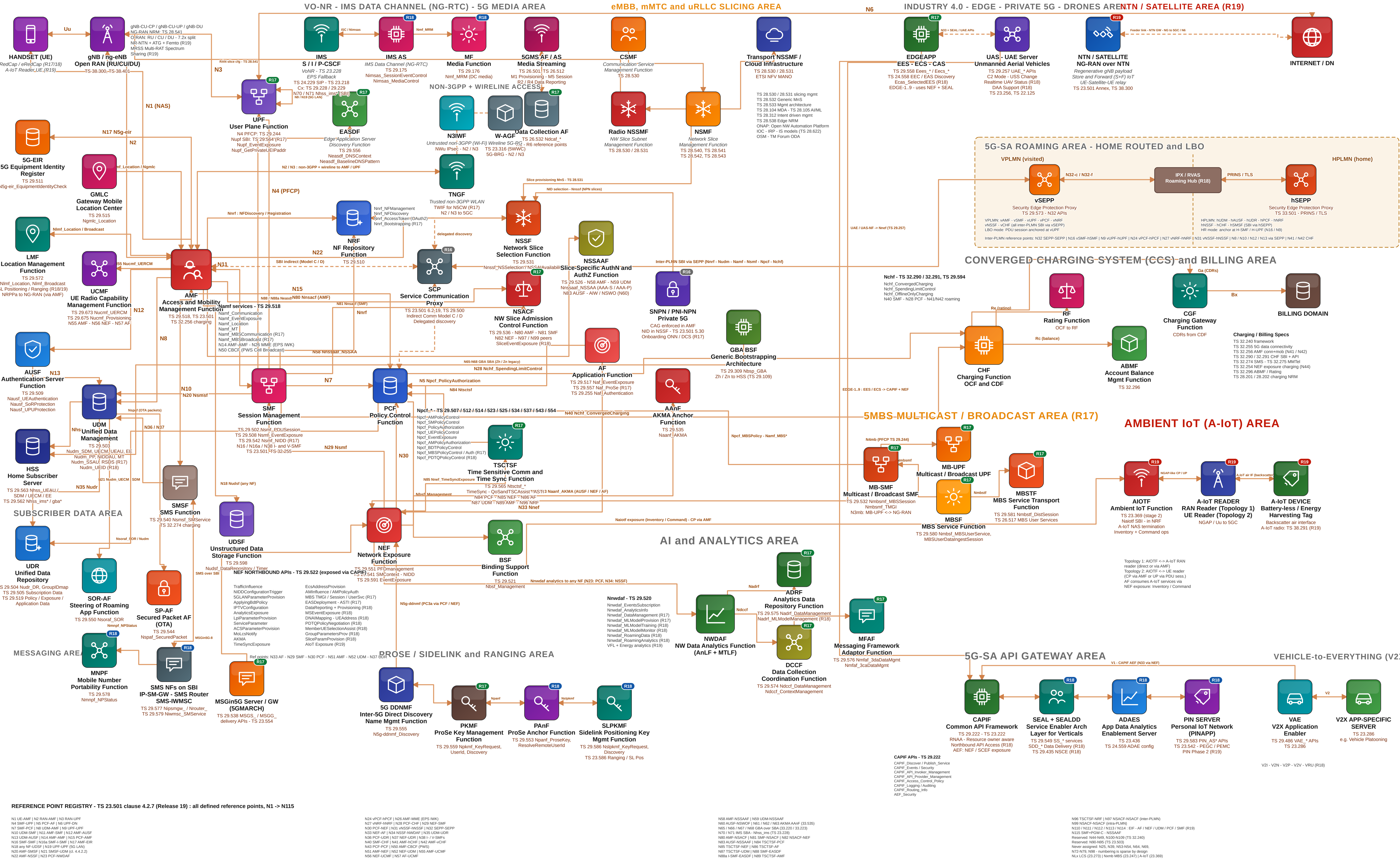


5G-Advanced Protocols and Node Capabilities - Release 19

SBA : HTTP/2 + OpenAPI 3.0 - TLS 1.3 - OAuth2
Aligned to 3GPP Force 5G APIs - REL-19 branch



RELEASE 19 HIGHLIGHTS

Ambient IoT - AIOTF (TS 23.369) | NTN Ph3 : Store and Forward + regenerative gNB payload | AI/ML Ph2 : VFL + energy analytics (TS 23.288) | XRM Ph2 | Energy Efficiency / Saving
Edge Ph3 | Ranging / Sidelink Positioning Ph2 (SL-PKMF) | PIN Ph2 | MPS Ph3 | MRSS (RAN) | RedCap evolution | IMS Data Channel / NG-RTC ecosystem

R16 Baseline (R15/R16) R17 New in Release 17 R18 New in Release 18 R19 New in Release 19

All Nxxx services ride the common SBI bus (HTTP/2 + JSON, OpenAPI 3.0). Reference points per TS 23.501 / 23.502 / 23.503. Charging per TS 32.240 series. Exposure northbound via NEF + CAPIF.

PRAJITH PARAN

mails@prajith.com

Rev : Release 19 - July 2026 (updated from R16/17/18 - 28 July 2022)

Sources : 3GPP Force 5G APIs (REL-19) - TS 23.501 / 23.502 / 23.503 - mirror github.com/degree5G/5G APIs